

Sedimentary Rocks	Metamorphic Rocks
Limestone	Marble
Sandstone	Quartzite
Shale/Clay	Slate, Phyllite, Schist,
Coal	Diamond or Graphite

Original Rocks	Metamorphic Rocks
Sandstone	Quartzite
Limestone	Marble
Shale and mudstone	Slate
Granite	Gneiss
Coal	Graphite coal
Clay	Slate

WEATHERING

- It is the process of disintegration or decomposition of rocks in Situ by natural agents. It is a static process.

There are three types of weathering as given below

- Physical Weathering** (or mechanical weathering)
 - (a) It involves rock disintegration without any change in the chemical constituents of the rocks.
 - (b) The factors responsible for physical weathering are temperature change, crystallisation of water into ice and pressure release mechanism.
- Chemical Weathering** It involves decomposition due to chemical changes. There are various chemical processes, which cause chemical weathering such as Solution, Oxidation, Carbonation, Hydration, Hydrolysis and Chelation.
- Biological Weathering** Plants and animals including man largely control it. *It is divided into three types*
 - (a) Faunal weathering
 - (b) Floral weathering
 - (c) Anthropogenic weathering

Erosion

It involves removal of rock material and then its transportation. Therefore, erosion is performed by mobile agents such as streams, wind, waves and underground water.

LANDFORMS

- A landform is a feature on the Earth's surface. It may be either the terrain, mountains, hills, plateaus or plains.
- Minor landforms include buttes, canyons, valleys and basins etc.

- Tectonic plate movement under the Earth can create landforms by pushing up mountains and hills.

Major classification of landforms on the basis of their process of origin is given as under :

Fluvial Landforms

A fluvial landform is a feature resulting from the movement of water on the Earth's surface. Flowing water is the most important force shaping the Earth.

- Erosional Landforms** Erosional landforms made by river are u-shaped valley, potholes, plunge pools, meanders, river terraces etc.
- Depositional Landforms** Depositional landforms made by river are alluvial fans, deltas, floodplains, natural levees, point bars etc.

Karst Landforms

Any limestone or dolomite region showing typical landforms produced by the action of ground water through the processes of solution and deposition is called Karst Topography.

- Erosional Landforms** Pools, sinkholes, lapies and limestone pavement, caves are important erosional landforms made by underground water.
- Depositional Landforms** Stalactites, stalagmites, pillars etc are important depositional landforms of underground water.

Glacial Landforms

The movement of glaciers is slow unlike the water flow. The movement could be a few centimetres to a few metres a day or even less. Glaciers move basically because of the force of gravity.

- Erosional Landforms** Cirque/Corrie U-shaped valley, horns, serrated ridges, glacial troughs, hanging valleys etc are important erosional landforms made by glaciers.
- Depositional Landforms** Moraines, eskers, outwash plains, drumlins etc are depositional landforms made by glaciers.

Coastal Landforms

Coastal landforms are formed by erosion or deposition through sea waves, aided by currents tides and storms in coastal areas.

- Erosional Landforms** Cliffs, terraces, caves, stacks etc are important landforms made by oceanic waves.
- Depositional Landforms** Beaches, dunes, bars, barriers, spits etc are important depositional landforms made by sea and oceanic waves.

Aeolian Landforms

Wind is one of the most dominant agents in hot deserts. The wind action creates a number of interesting erosional land depositional features in the deserts.

Aeolian landforms are regions where erosion and deposition by wind are dominant geomorphic forces in shaping the landform. They are formed in hot deserts, coastal zones and fields.

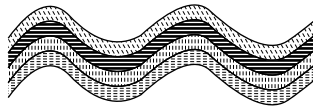
- Erosional Landforms** Pediments, pediplains, playas, deflation hollows, mushroom rock, table rock, pedestal rocks etc are erosional landforms in deserts.
- Depositional Landforms** Sand dunes, barchans, seifs etc are depositional landforms made by wind.

MOUNTAINS

Based on their mode of formation four main types of mountains can be distinguished.

Fold Mountains

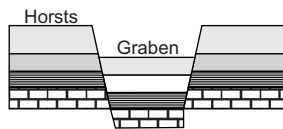
- It is formed due to the compressive forces generated by endogenetic forces (earthquake, landslide etc.)
- Example of Fold mountains are Himalayas, Alps, Andes, Rockies, Atlas etc.
- *On the basis of age, fold mountains are grouped into*
 - **Young/New Fold Mountains** It came into existence after the continental drift, e.g. Himalayas, Andes, Rockies, Alps. Himalayas are regarded as the youngest mountains in the world.
 - **Old Fold Mountains** They belong to pre-drift era, e.g. Pennines (Europe), Appalachians (US), Aravallis (India) etc.



Fold Mountains

Block Mountains

- It is formed when a huge block of Earth's crust may be raised or lowered due to tectonic activities. The land between the two parallel faults either rises forming block mountains or horsts or subsides into a depression termed as Rift valley or Graben.
- Examples of Block mountain are Vindhya and Satpura in India, the Vosges in France and Black forest in Germany (through which Rhine river flows).



Block Mountains

Volcanic Mountains

- They are formed due to the accumulation of volcanic material. They are also known as **Mountains of Accumulation**.
- Examples of Volcanic mountain are Mt Fuji (Japan), Cotopaxi in Andes, Vesuvius and Etna in Italy, Mt Mayon (Philippines) etc.

Residual or Dissected Mountain

- They are formed as a result of erosion of plateaus and high planes by various agents of erosion.
- Examples of Residual or Dissected mountain are Catskill mountains of New York, Sierras of Spain, Girnar and Rajmahal of India.

Mountain Range	Location	Highest Point
Himalayas	Asia	Mount Everest
Karakoram	Asia	K2
Hindu Kush	Asia	Tirich Mir
Andes	South America	Aconcagua
Alps	Europe	Mount Blanc
Rockies	North America	Mount Elbert
Urals	Asia/Europe	Mount Narodnaya
Alaska Range	North America	Denali (Mount McKinley)
Transantarctic Range	Antarctica	Mount Kirkpatrick
Caucasus	Europe/Asia	Mount Elbrus
Tien Shan	Asia	Jengish Chokusu
Great Dividing Range	Australia	Mount Kosciuszko

PLATEAUS

Tabular upland having relief of more than 500 feet may be defined as plateau. Tibetan plateau (5000 m) is the highest plateau in the world.

Types of Plateau

According to their mode of formation and their physical appearance, plateau may be grouped into the following types :

Tectonic Plateau

- These are formed by Earth movements, which cause uplift and are normally of a considerable size and fairly uniform altitude.
 - When plateau are enclosed by Fold mountains, they are known as **Intermontane plateau**.
 - Examples of Tectonic plateau are Tibetan plateau between the Himalayas and the Kunlun and the Bolivian plateau between two ranges of the world.

Volcanic Plateau

These are formed by accumulation of lava. e.g. Deccan Plateau (India).

Dissected Plateau

- Through the continual process of weathering and erosion by running water, ice and winds, high extensive plateau are gradually worn down and their surface made irregular.
- The Scottish Highlands is a good example of dissected plateau.

PLAINS

A relatively low-lying and flat land surface with least difference between its highest and lowest points is called a **plain**. The plains are divided into structural, erosional and depositional plains.

Types of Plain

There are three types of plain as follow :

- i. **Structural Plain** Formed due to the uplift of a part of the sea floor e.g. the great plain of USA.
- ii. **Erosional Plain** Formed when the elevated tract of land is worn down to a plain by the process of erosion. e.g. Plain of North Canada.
- iii. **Depositional Plain** Formed by filling up of sediments into depressions along the foothills, lakes and seas e.g. Indo Ganga Plain.

ISLANDS

An island is any piece of land that is surrounded by water. A grouping of geographically or geologically related islands is called an Archipelago. The various types of islands are named according to the way they were formed.

Types of Island

Types of island are as follow

Continental Island

- It is simply an island that rests on the continental shelf. Because of this, these islands are always quite close to a given continent. Also, the water level around a continental island is very shallow, typically less than 600 feet.
- Canadian Island of Newfoundland is a continental island, as is Greenland. Great Britain is the largest continental island in Europe.

Volcanic Island

They are formed when volcanoes deep in the ocean rise above the water's surface.

Coral Island

- It is an island that forms a ring and partially or totally encloses a shallow body of water or lagoon. Coral islands, which are large collections of coral a top rock (usually volcanic) in the ocean.
- Lakshadweep island in Indian ocean is best example of Coral islands.

Alluvial Island

- They are formed by the outward flow of water depositing silt and gravel at the mouth of a river.
- New Moore island at the mouth of Ganga river in West Bengal is known as alluvial island.

Barrier Island

They are formed by waves depositing sand on the shallow ocean bottom as they crash on the shoreline.

Tectonic Island

- They are formed by the result of tectonic activity. By the tectonic movement long island arcs (archipelagoes) are formed.
- Indonesia, Philippines, Japan groups of island are best examples of tectonic island

LAKES

A lake is a body of water surrounded by land from all the sides. A lake can be fed and drained by rivers and streams.

Types of Lake

Lakes can be classified as follow :

Tarn Lake

Tarn lakes (also often classified as Alpine lakes) are usually associated with glacial regions areas that have been glaciated in the past. These follows form as the ice scours out the side of a hill, creating what is called a cirque.

Rift Valley Lake

- Rift valley lakes occur in areas where rock is pulling apart. As a narrow wedge of land drops from the movement, a long, narrow lake often forms.
- The Rift valley, which cuts through Africa and Asia, contains the largest group of such lakes, including lake Nyasa.

Crater Lake

- Crater lakes form at a volcanic peak. When an extinct volcano collapses, it often leaves a somewhat circular depression; water from rain and runoff eventually fills the lake.
- Crater lake in Oregon is a good example of such a lake.

Deflation Lake

- Deflation lakes usually occur in deserts as the wind blows out depressions in the sand. If the hole reaches the local groundwater table, it can create an oasis.
- Such features dot the Sahara desert in Africa which are locally called **Oasis**.

Oxbow Lake

- Oxbow lakes are created by a curved river meander that was cut off by sediment or other means, creating a bowed shaped lake.
- The Mississippi river shows evidence of past Oxbow lakes.

Artificial Lake

- Artificial lakes are those usually formed by the damming of a river.
- These lakes are usually created to regulate the flow of a seasonal flooding river and many times offer a constant supply of potable (drinkable) water or generated electricity.

Lakes	Countries	Area (km ²)
Caspian Sea	Kazakhstan, Turkmenistan, Iran, Azerbaijan, Russia	436000
Superior	Canada, United States	82414
Victoria	Uganda, Kenya, Tanzania	69485
Huron	Canada, United States	59,600
Michigan	United States	58000
Tanganyika	Burundi, Tanzania, Zambia	32893
Baikal	Russia	31500
Great Bear Lake	Canada	31080
Malanei Malawi	Tanzania, Malawi, Mozambique	30044
Great Slave Lake	Canada	28930
Balkhash	Kazakhstan	18428
Ladoga	Russia	18130
Nicaragua	Nicaragua	8001
Great Salt Lake	United States	4662

ATMOSPHERE

- Atmosphere is a thick gaseous envelope which surrounds the Earth from all sides and it is attached to Earth through the force of gravitation.
- **Extent of Atmosphere** Almost 97% of effective atmosphere confines within the height 29 km of the Earth's surface. So, the vertical distribution of the atmosphere is not uniform and even not homogeneous. It extends between 16-29000 km from the sea level.

Composition

- The atmosphere is mainly composed of gases, water vapour and particulates. The main gases found in atmosphere are nitrogen (78%), oxygen (21%), argon (0.33%), carbon dioxide (0.03%), neon, helium, ozone, hydrogen etc.
- Besides these gases, water vapour, dust particles and other particulate matter are also present. Ozone gas is found in a belt lying at an altitude of around 20-25 km. The amount of water vapour present in the atmosphere varies from region to region between 0-4% by volume.

Layers of Atmosphere

Troposphere

- It extends upto 16 km from the Earth's surface. Thickness varies from 8 km at the poles to 18 km at the equator. At every 165 m, there is a drop of 1°C (or 6.4°C per km). This is called **normal lapse rate of temperature**.

- **Tropopause** separates troposphere from stratosphere. Troposphere accounts for practically the entire water vapour, all dust particles and most of the carbon dioxide contained in the atmosphere. Due to this all weather phenomena such as condensation, precipitation and storms etc occur in the troposphere only.

TEMPERATURE INVERSION

- It refers to reversal of normal behaviour of temperature in troposphere such that a layer of cool air at the surface is overlain by a layer of warmer air.
- Inversion plays an important role in determining cloud forms, precipitation and visibility. Long nights with clear skies and, calm and stable air are ideal conditions for temperature inversion.

Stratosphere

- The stratosphere extends upto about 50 km, where stratopause separates it from the mesosphere.
- In stratosphere, the temperature increases with increase in height. This phenomenon is known as Temperature Inversion.
- The temperature rises in this layer from about -60°C at the tropopause to 0°C at stratopause. The part of the stratosphere, in which there is a concentration of ozone is often called Ozonosphere. It absorbs ultraviolet radiation, which is harmful for us.
- Stratosphere is free from dust particles and also from atmospheric turbulences. Hence, this layer is considered ideal for flying of jet aircrafts.

Mesosphere

- Mesosphere extends above the stratopause upto a height of about 80 km. In this layer, the temperature decreases with height like in the troposphere and it falls from about 0°C at its base to about -100°C at 80 km height. It is considered the coldest layer of the atmosphere.
- The upper limit of the mesosphere is marked by the **Mesopause**, a transitional layer separating it from the ionosphere.

Ionosphere

- Ionosphere is located above the mesosphere and extends upto about 600 km. This layer is called as ionosphere because it contains electrically charged ions that reflect the radio waves back to the Earth, thus making radio communication possible.
- Absorption of solar radiation by ionised particles cause an increase in temperature with increasing height in the ionosphere.
- Due to large concentration of ionised particles in this layer the ionosphere acts as a protective layer against meteorites, that are burnt in this layer.

Thermosphere

- The zone between the 85 km and 400 km above the surface is often called Thermosphere. In this layer, the temperature increases with increasing altitude. The upper limit of the **thermosphere**, the **thermopause** is generally taken at an altitude of about 600 km.
- The day temperature at 600 km altitude exceed 1400°C while night temperature remain about 225°C .
- The upper part of the thermosphere contains only the lighter gases like helium and hydrogen.

Exosphere and Magnetosphere

- The outermost part of the atmosphere of the Earth is called Exosphere. This zone of the atmosphere extends upto a height of about 900 km.
- The upper limit of the exosphere is uncertain as this layer acts as a transitional layer between the Earth's atmosphere and the space. The outer part of the exosphere is called Magnetosphere. The presence of air is extremely rare in this layer.

Insolation and Solar Radiation

- Sun is the main source of energy on Earth. Energy from Sun is received in the form of radiation. Solar radiation that is intercepted by Earth is known as Insolation.

The amount of insolation depends on following factors :

- The area and nature of the surface.
- The inclination of the rays of the Sun.
- Distance between the Earth and the Sun.
- Length of the day.
- The transparency of the atmosphere.
- As the angle of the Sun rays decreases poleward, the amount of insolation received also decreases in that direction.
- The Earth's surface does not absorb all the energy that it receives. The proportion of the solar radiation reflected from the surface is called Albedo.
- On an average, insolation is highest near the equator, marginally lower at the tropic and lowest at the poles.

Heat Budget of the Earth

The Earth receives energy continuously from the Sun, its temperature is almost constant except the long-term climate changes. This is because the atmosphere loses an amount of heat equal to the gain through insolation. This mechanism of maintaining the same temperature by the atmosphere is called the Heat Budget or Heat Balance.

The heated surface of the Earth radiates energy in the form of long waves. This process is called Terrestrial Radiation.

Atmospheric Pressure

Air is an extremely compressible gas having its own weight. The pressure exerted by air due to its weight is called atmospheric pressure on the Earth's surface. Atmospheric pressure is neither same for all the regions nor same for one region all the time.

Factor Affecting Atmospheric Pressure

Atmospheric pressure is affected by various factors such as altitude, temperature and Earth's rotation.

Altitude Air pressure increases, when air descends due to the decrease in volume. When air rises its volume increases and the outward pressure of its molecule is spread over a larger area and its pressure decreases.

Temperature The pressure of air rises, when its temperature falls. Low temperatures at the poles cause the air to contract—high pressure develops; whereas the high temperature along the equator cause the air to expand—low pressure develops.

Pressure Belts

Earth can be divided into regions of broader pressure belts, even though it has regional anomalies or deviations.

Equatorial Low Pressure Belt (or Doldrums)

- It is located on either side of the geographical equator in a zone extending between 5°N and 5°S . Its location is not stationary and there is a seasonal drift of this belt with the Northward (summer solstice) and Southward (winter solstice) migration of the Sun.
- It is thermally induced because of the intense heating of the ground surface by the almost vertical Sun rays.
- It represents the zone of convergence of North East and South East trade winds. Due to the frequent calm conditions this belt is called as a **belt of calm** or **doldrums**.

Inter Tropical Convergence Zone (ITCZ)

- The ITCZ is a low pressure zone located at the equator where trade winds converge and so, it is a zone where air tends to ascend.
- In July, the ITCZ is located around 20°N - 25°N latitudes cover the gangetic plain), sometimes called the Monsoon Trough. This monsoon trough encourages the development of thermal low over North and North-West India.
- Due to the shift of ITCZ, the trade winds of the Southern hemisphere cross the equator between 40° and 60° E longitudes and start blowing from South-West to North-East due to the Coriolis force. It becomes South-West monsoon.
- In winter, the ITCZ moves Southward and so the reversal of winds from North-East to South and South-West, takes place. They are called North-East monsoons.